

ALEKSANDAR STOJKOSKI

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SUMMARY

Third-year Mechanical Engineering (Honours) student with experience delivering client-focused mechanical and structural design solutions. Skilled in stakeholder engagement, finite element analysis and engineering decision-making, with experience translating complex technical problems into practical, evidence-based solutions.

EDUCATION

Bachelor of Engineering (Honours) – Mechanical

Expected Completion 2027

Deakin University, Burwood

Relevant coursework: Control Systems Engineering, Dynamics of Machines, Thermo-Fluid Systems, Product Design

ENGINEERING PROJECTS

Design Engineering Project - Universal Chain Guide Attachment (SpiderGuide)

2026

- Designed the "SpiderGuide", a universal chain guide attachment for electric chain hoists, addressing a chain-spooling issue identified by Melbourne Park and delivering a practical engineering solution in response to a real client brief.
- Translated client and stakeholder requirements into measurable engineering specifications by benchmarking five commercial products, resulting in a tool-free, non-invasive design weighing less than 5 kg and suitable for client evaluation.
- Led structural validation of the final design using ABAQUS finite element analysis (static, frequency and buckling), confirming structural integrity under operating loads and identifying the clamp bracket as the critical component (minimum factor of safety: 2.37), enabling targeted design optimisation.

Formula SAE Team Establishment Initiative - Deakin University

2026-Present

- Co-led the proposal for Deakin Burwood's first formula SAE program, transforming a conceptual initiative into a structured implementation plan through stakeholder consultation and strategic planning.
- Facilitated collaboration between academic staff and university leadership to align competing stakeholder priorities, resource availability, and long-term program objectives.
- Produced a business case and executive presentation for senior engineering leadership, evaluating program feasibility and demonstrating its value to student outcomes, industry engagement, and university growth.

Structural Design Project - Cyclist & Pedestrian Bridge

2025

- Developed the preliminary structural design of a cable-stayed cyclist and pedestrian bridge for the City of Stonnington, analysing the main girder, A-frame, stay cables and footing using SPACE GASS and independent hand calculations to verify structural performance.
- Responded to revised geotechnical constraints by redesigning the footing for a reduced allowable bearing pressure of 100 kPa, maintaining compliance with bearing ($FoS \geq 3.0$) and sliding/overturning ($FoS \geq 1.5$) requirements without redesigning the entire structure.

SKILLS

Technical Skills

Structural & Analysis

- SPACE GASS
- ABAQUS FEA

CAD

- Fusion 360

Programming

- Python
- MATLAB
- AI-assisted workflows (Claude Code)

Soft Skills

- Clear communication and technical presentation
- Analytical problem-solving
- Teamwork and collaboration
- Stakeholder engagement
- Time management and adaptability

PROFESSIONAL EXPERIENCE

Retail Athlete (Sales Assistant) – Nike:

2022 – Present

- Delivered customer service in a fast-paced retail environment while consistently balancing full-time engineering studies, collaborating across teams to achieve daily sales and operational targets.

VOLUNTEERING & ACTIVITIES

Deakin Engineering Ambassador - Burwood Campus:

2024 – Present

- Selected by the Head of Engineering in first year to represent Deakin's inaugural Burwood engineering cohort, presenting student projects to politicians, council representatives and industry stakeholders at a major campus promotion event.
- Supporting ongoing advocacy for engineering at Deakin through orientation and outreach events.

Track Marshal - Preston Motorcycle Club, Broadford:

2024 – Present

- Safety-critical trackside role requiring sustained situational awareness and rapid incident response.